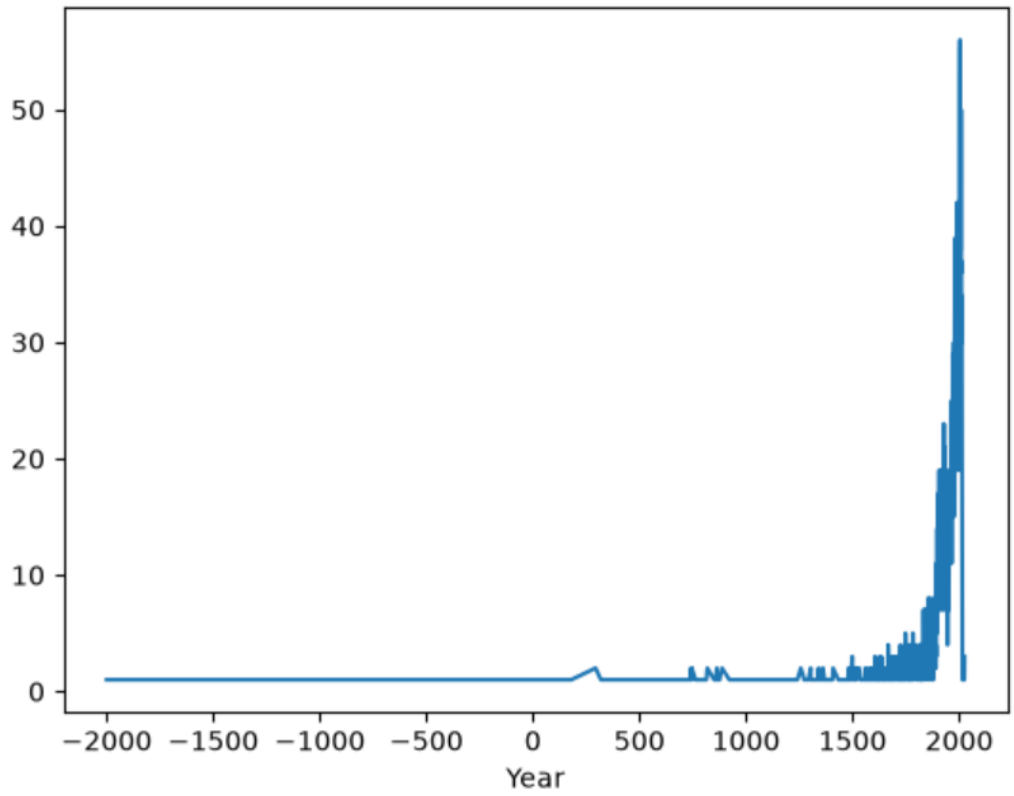
1.Significant earthquakes since 2150 B.C.

1.1

```
Country
CHINA           2106526.0
TURKEY          1008867.0
IRAN             761656.0
SYRIA            488326.0
ITALY            417297.0
HAITI            323782.0
JAPAN            319510.0
AZERBAIJAN       319251.0
INDONESIA        283485.0
ARMENIA          175000.0
PAKISTAN         143714.0
ECUADOR          134445.0
TURKMENISTAN     110412.0
PERU              97009.0
GREECE            83658.0
PORTUGAL          82572.0
IRAQ              70200.0
CHILE             70175.0
INDIA             63413.0
ISRAEL            58368.0
Name: Total Deaths, dtype: float64
```

1.2 趋势：震级>3 地震总数近些年急剧上升，原因可能是过去技术不发达，监测不全



1.3

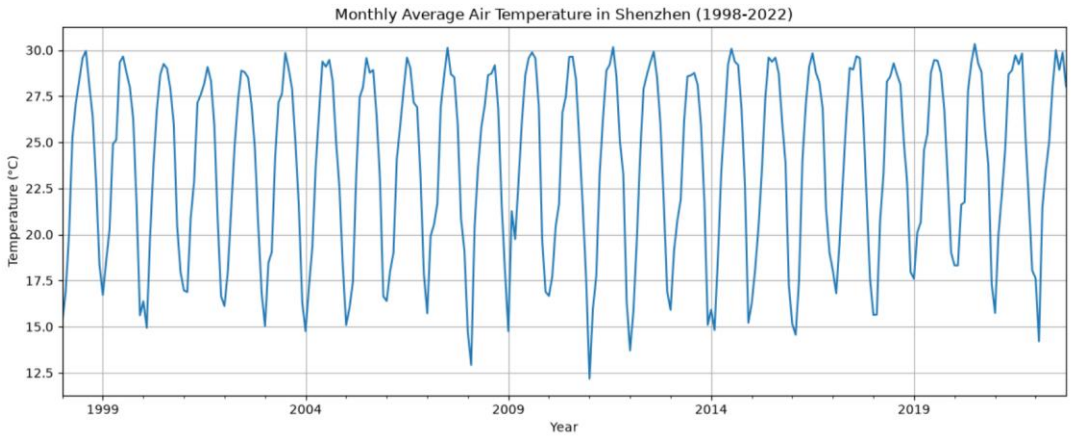
	Country	Total_Earthquake	Largest_Ms	Largest_Date	Largest_Location
15	CHINA	626	8.6	1920.0-12.0-16.0	CHINA: GANSU PROVINCE, SHANXI PROVINCE
34	JAPAN	429	8.6	869.0-7.0-13.0	JAPAN: SANRIKU
74	INDONESIA	427	8.8	2004.0-12.0-26.0	INDONESIA: SUMATRA: ACEH: OFF WEST COAST
8	IRAN	385	7.9	856.0-12.0-22.0	IRAN: DAMGHAN, QUMIS
10	TURKEY	359	7.8	1939.0-12.0-26.0	TURKEY: ERZINCAN
...	...	...	...	...	...
155	BURUNDI	1	unknown magnitude	unknown date	unknown location
161	ZAMBIA	1	unknown magnitude	unknown date	unknown location
160	MADAGASCAR	1	unknown magnitude	unknown date	unknown location
162	COMOROS	1	unknown magnitude	unknown date	unknown location
0	NaN	0	unknown magnitude	unknown date	unknown location

163 rows × 5 columns

2. Air temperature in Shenzhen 1998-2022

如何过滤数据：删除温度缺失值，将温度转换为摄氏度，保留质量代码为"0", "1", "4", "5", "9"的数据

1998-2022 年月平均气温年际波动存在周期性，总体成缓慢上升趋势。



3. Global collection of hurricanes

3.1

```
SID            NAME
1950224N17305  ABLE       95
1950234N11334  CHARLIE    95
1950244N20277  EASY       95
1950251N16319  FOX        95
1950270N25309  GEORGE     95
1951224N12316  CHARLIE    95
1951244N13328  EASY       95
1951246N13340  FOX        95
1952244N15306  BAKER      95
1952268N20291  CHARLIE    95
Name: WMO_WIND, dtype: str
```

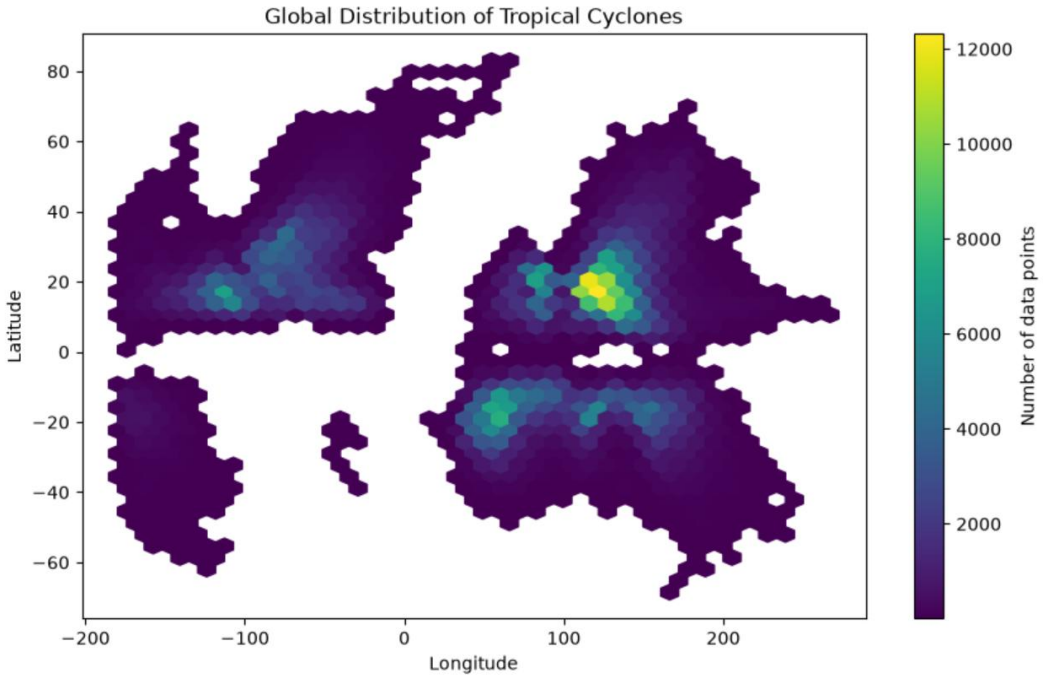
3.2

```
SID            NAME
1950224N17305  ABLE        95
1950234N11334  CHARLIE     95
1950244N20277  EASY        95
1950251N16319  FOX         95
1950270N25309  GEORGE      95
1951224N12316  CHARLIE     95
1951244N13328  EASY        95
1951246N13340  FOX         95
1952244N15306  BAKER       95
1952268N20291  CHARLIE     95
1952295N11282  FOX         95
1953240N14340  CAROL       95
1953258N19296  EDNA        95
1953267N15282  FLORENCE    95
1954176N22266  ALICE       95
1954238N24285  CAROL       95
1954248N20295  EDNA        95
1954278N12302  HAZEL       95
1955215N15320  CONNIE      95
1955253N14319  IONE        95
Name: WMO_WIND, dtype: str
```

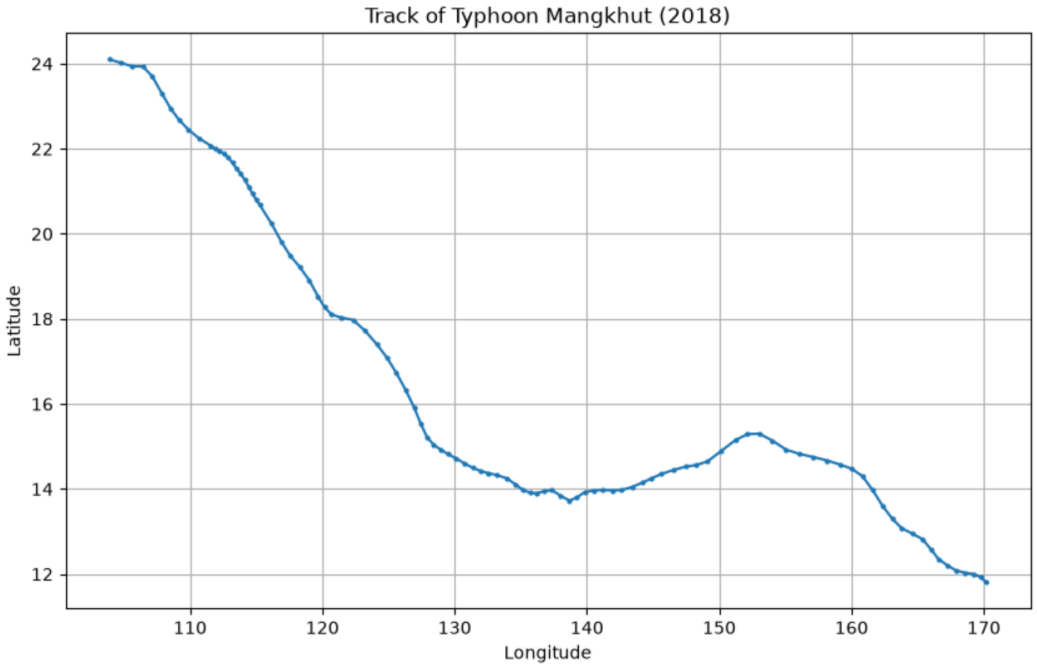
3.3

```
BASIN
WP    236576
SI    160668
SP     67119
EP     62412
NI     55401
SA       119
Name: count, dtype: int64
```

3.4


Global Distribution of Tropical Cyclones
Latitude
Longitude
Number of data points
80
60
40
20
0
−20
−40
−60
−200
−100
0
100
200
12000
10000
8000
6000
4000
2000

3.5


Track of Typhoon Mangkhut (2018)
Latitude
Longitude
24
22
20
18
16
14
12
110
120
130
140
150
160
170

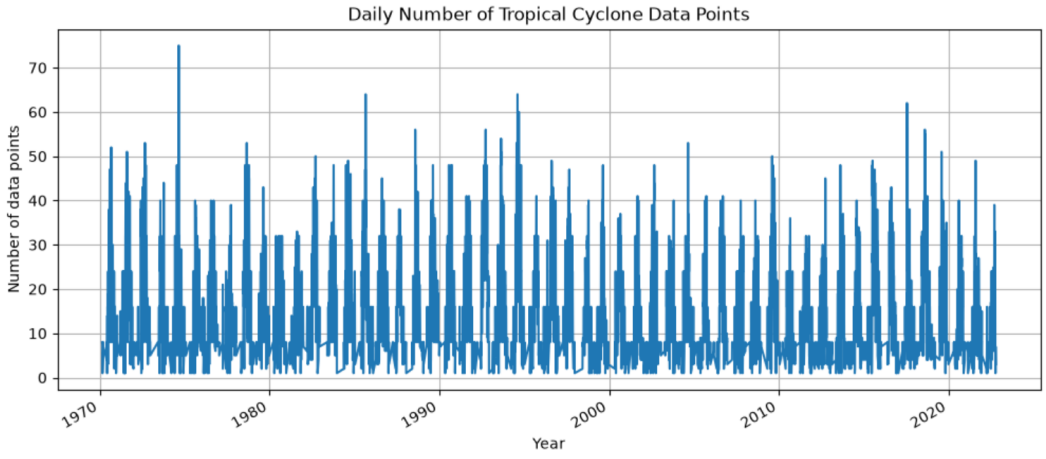
3.6

```
                SID  SEASON  NUMBER BASIN SUBBASIN    NAME  \
350393  1970050N07151    1970      22    WP        MM  NANCY
350394  1970050N07151    1970      22    WP        MM  NANCY
350395  1970050N07151    1970      22    WP        MM  NANCY
350396  1970050N07151    1970      22    WP        MM  NANCY
350397  1970050N07151    1970      22    WP        MM  NANCY

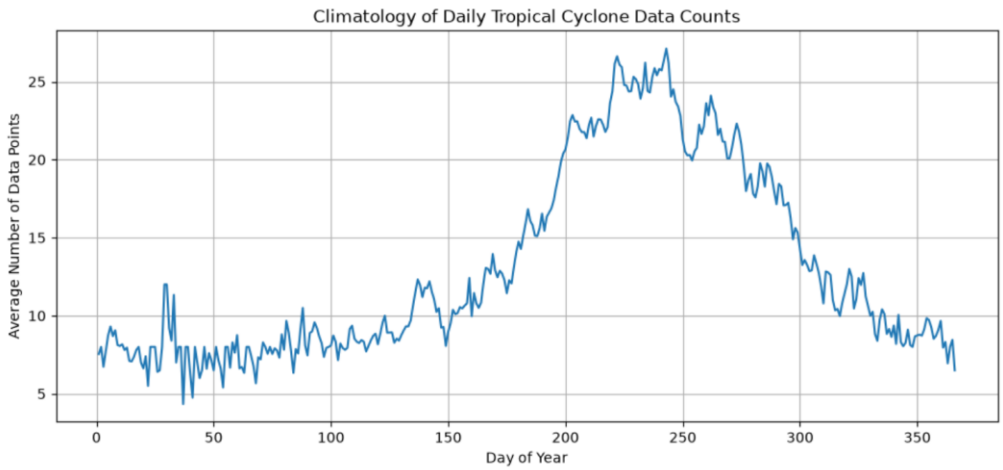
                ISO_TIME NATURE      LAT      LON WMO_WIND WMO_PRES  \
350393 1970-02-19 00:00:00     TS  7.00000  151.400              1006
350394 1970-02-19 03:00:00     TS  7.24752  151.205
350395 1970-02-19 06:00:00     TS  7.50000  151.000              1002
350396 1970-02-19 09:00:00     TS  7.75747  150.772
350397 1970-02-19 12:00:00     TS  8.00000  150.500               998

       WMO_AGENCY TRACK_TYPE  DIST2LAND LANDFALL           IFLAG
350393      tokyo       main       1088     1088  _OO_____________
350394                  main       1109     1109  _PP_____________
350395      tokyo       main       1143     1143  _OO_____________
350396                  main       1171     1168  _PP_____________
350397      tokyo       main       1182     1179  _OO_____________
```

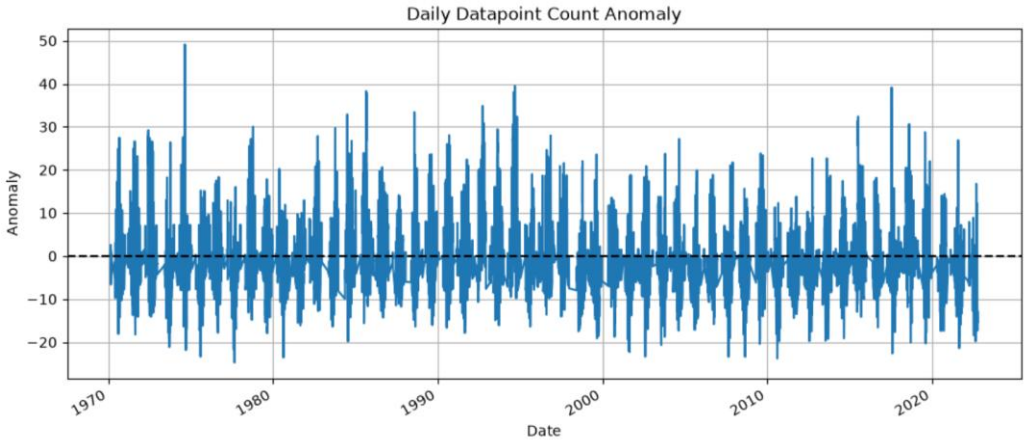
3.7



3.8

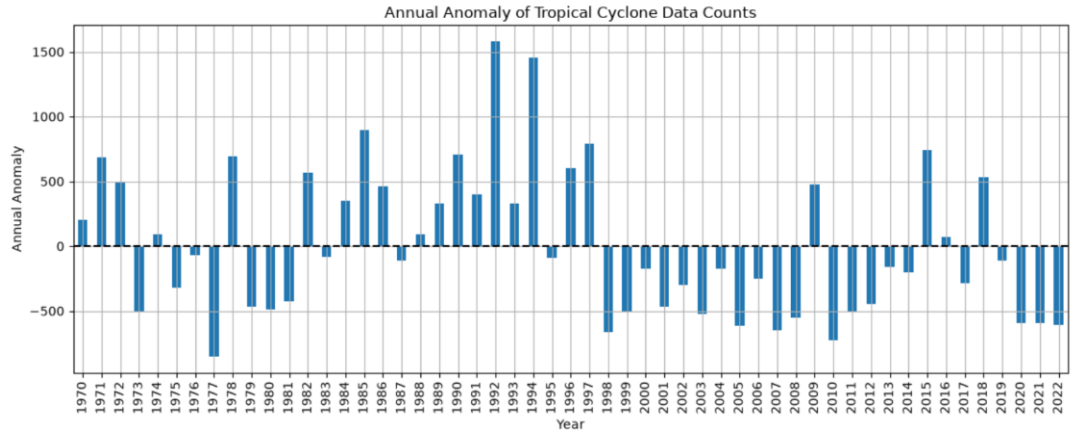


3.9



3.10 异常值最大的 5 个年份：1992，1994，1985，1997，2015

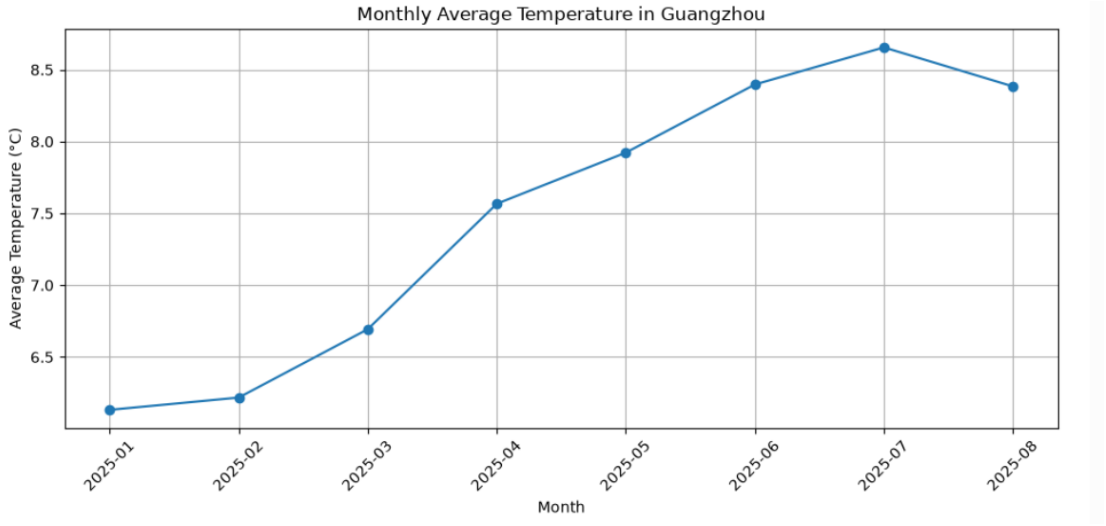
异常值最小的 5 个年份：1997，2010，1998，2007，2005



4. Explore a data set

```
<class 'pandas.DataFrame'>
RangeIndex: 236 entries, 0 to 235
Data columns (total 12 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   STATION          236 non-null    str
 1   NAME             236 non-null    str
 2   LATITUDE         236 non-null    float64
 3   LONGITUDE        236 non-null    float64
 4   ELEVATION        236 non-null    float64
 5   DATE             236 non-null    str
 6   TAVG             236 non-null    int64
 7   TAVG_ATTRIBUTES  236 non-null    str
 8   TMAX             105 non-null    float64
 9   TMAX_ATTRIBUTES  105 non-null    str
 10  TMIN             219 non-null    float64
 11  TMIN_ATTRIBUTES  219 non-null    str
dtypes: float64(5), int64(1), str(6)
memory usage: 32.0 KB
       STATION           NAME  LATITUDE  LONGITUDE  ELEVATION        DATE  \
0  CHM00059287  GUANGZHOU, CH    23.217    113.483       71.0  2025-01-01
1  CHM00059287  GUANGZHOU, CH    23.217    113.483       71.0  2025-01-02
2  CHM00059287  GUANGZHOU, CH    23.217    113.483       71.0  2025-01-03
3  CHM00059287  GUANGZHOU, CH    23.217    113.483       71.0  2025-01-04
4  CHM00059287  GUANGZHOU, CH    23.217    113.483       71.0  2025-01-05

   TAVG TAVG_ATTRIBUTES  TMAX TMAX_ATTRIBUTES  TMIN TMIN_ATTRIBUTES
0   6.2            H,,S   NaN             NaN  49.0            ,,S
1   6.5            H,,S   NaN             NaN  50.0            ,,S
2   6.2            H,,S  76.0            ,,S  45.0            ,,S
3   6.4            H,,S   NaN             NaN  44.0            ,,S
4   6.2            H,,S  74.0            ,,S  47.0            ,,S
数据总数： 236
缺失值数量：
DATE    0
TAVG    0
dtype: int64
```



平均气温： 7.48 °C
气温中位数： 7.8 °C
气温标准差： 1.1 °C
最高平均气温： 9.2 °C
最低平均气温： 5.0 °C
有效数据总数： 236